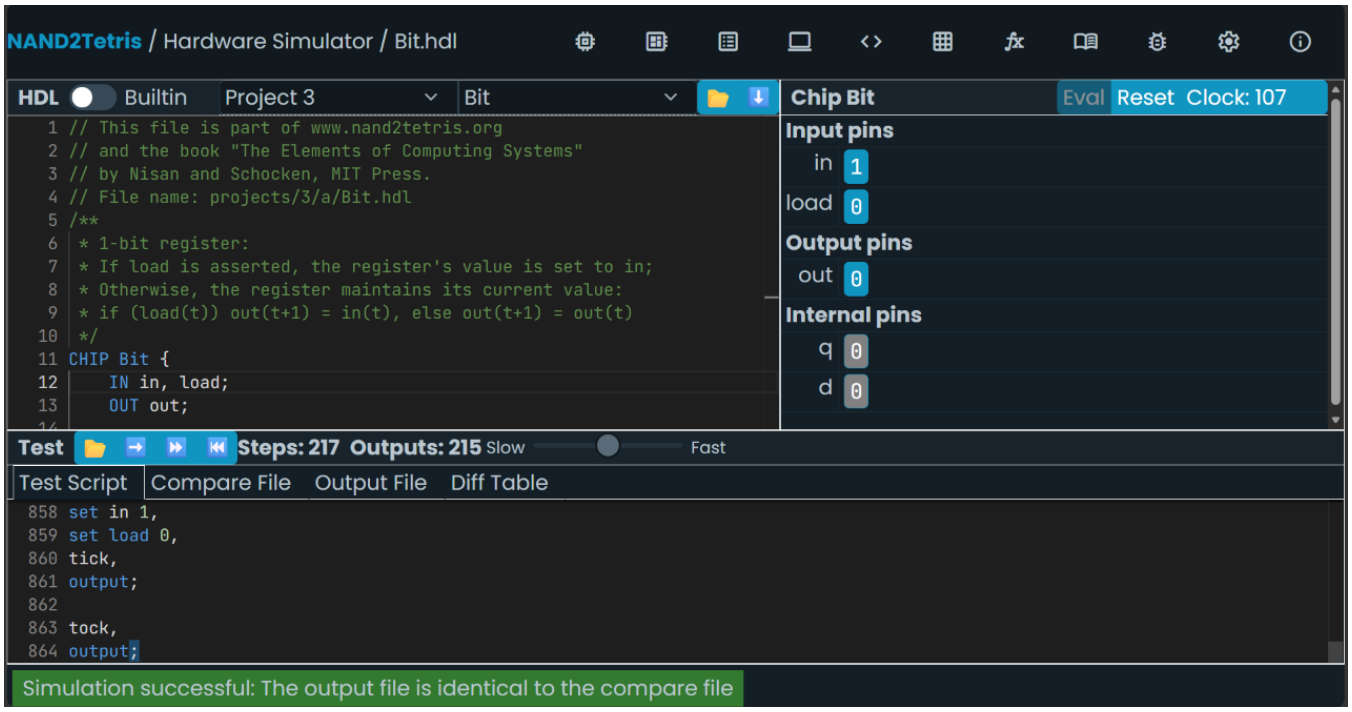
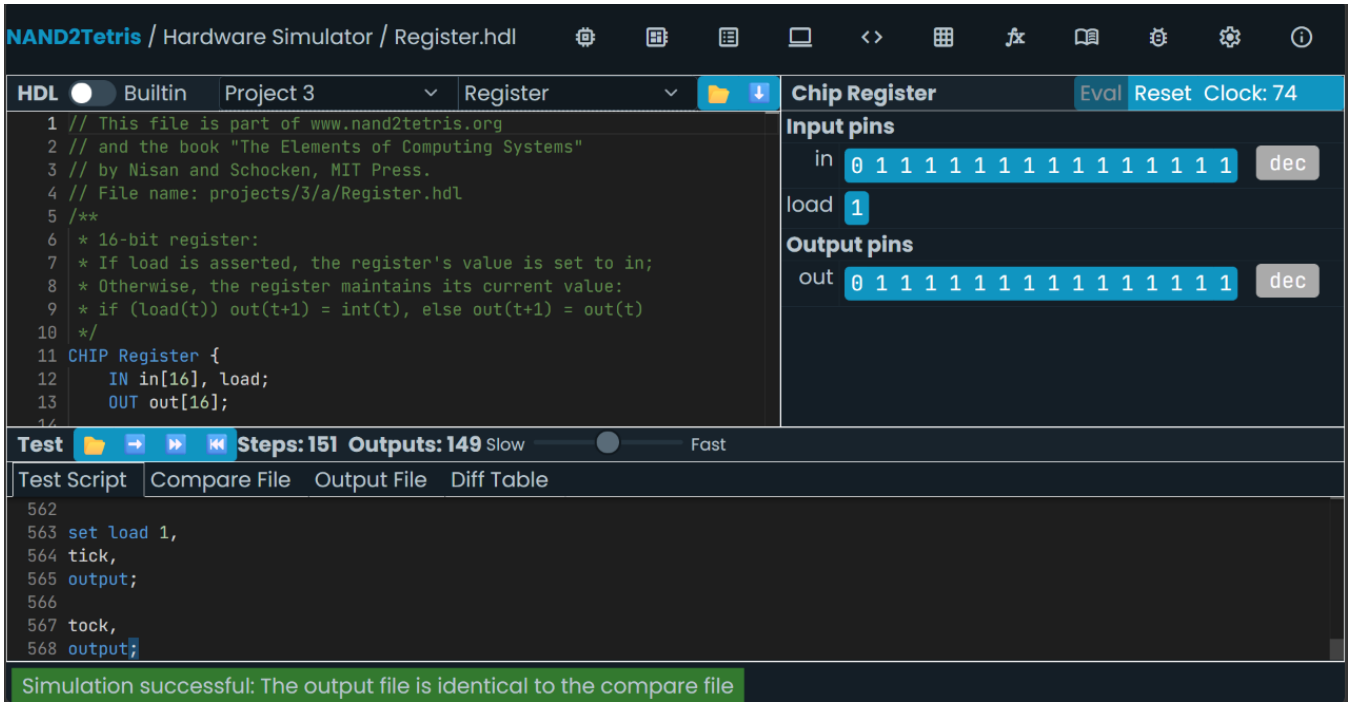
NAND2Tetris Project 3: Simulation Results

Hardware Simulator output for each chip. All simulations passed.

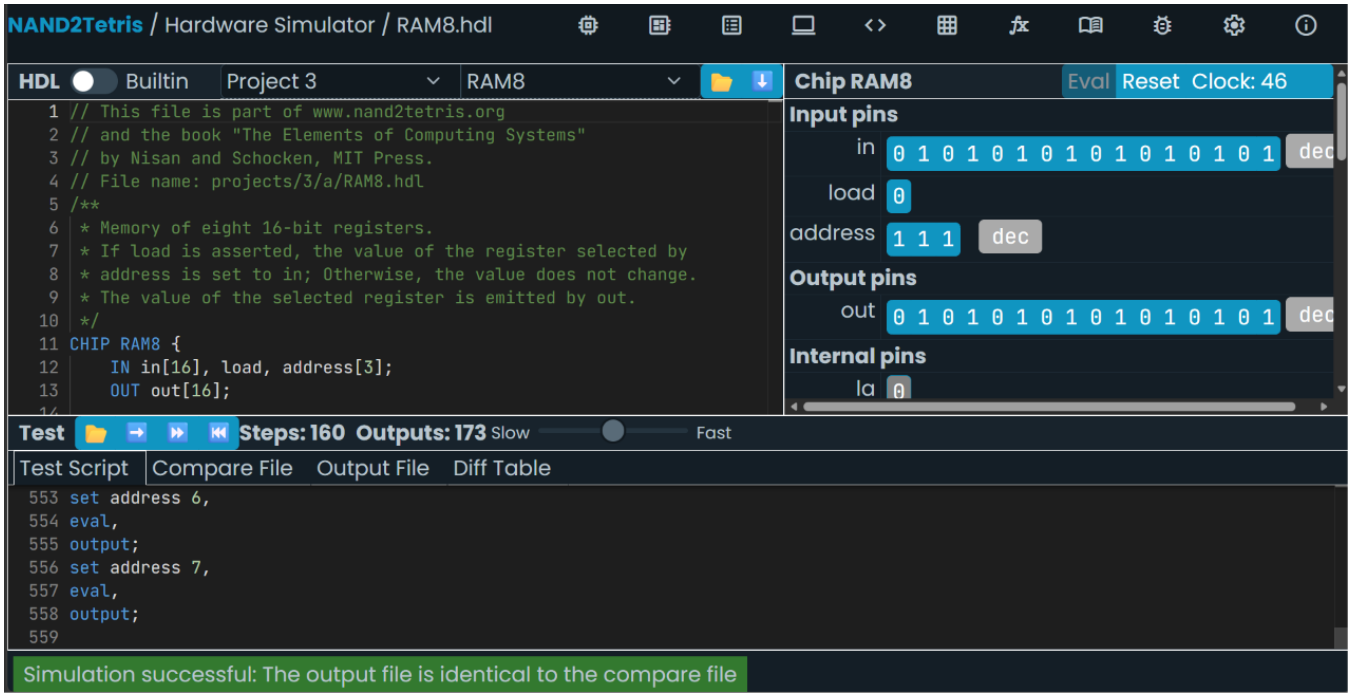
1. Bit



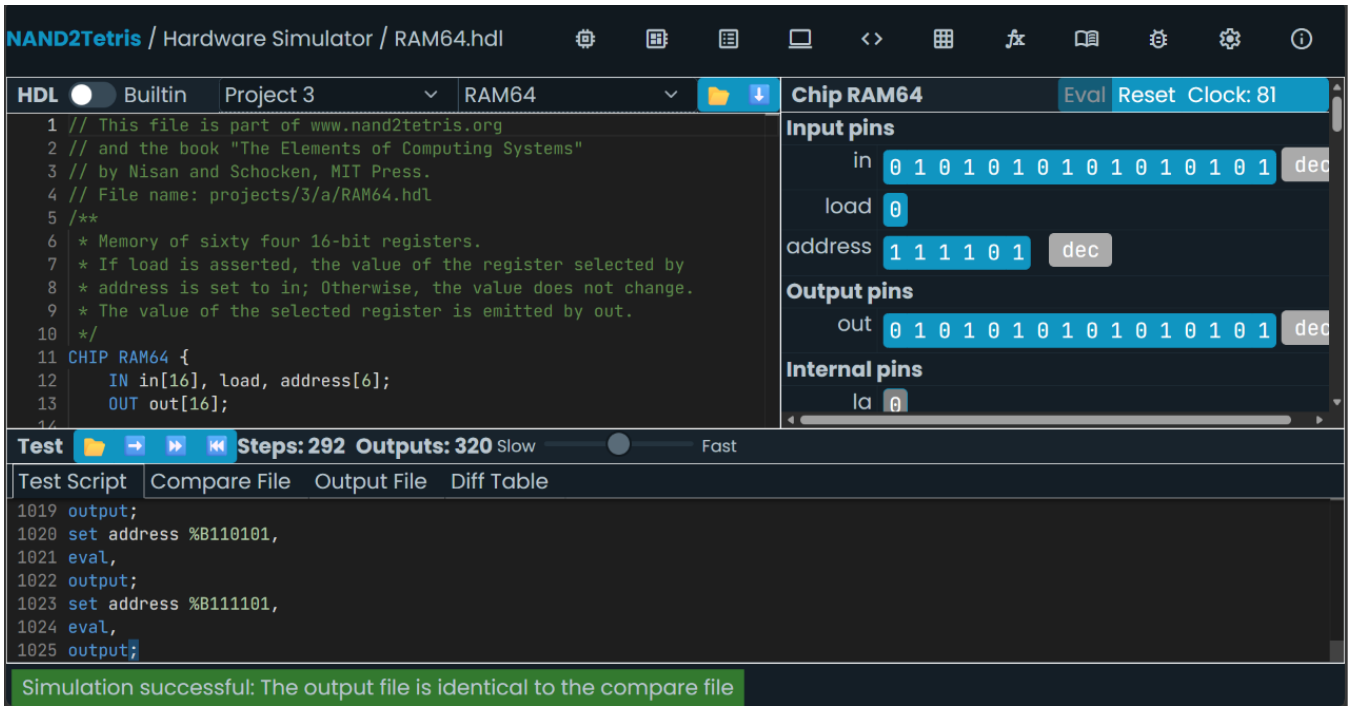
2. Register



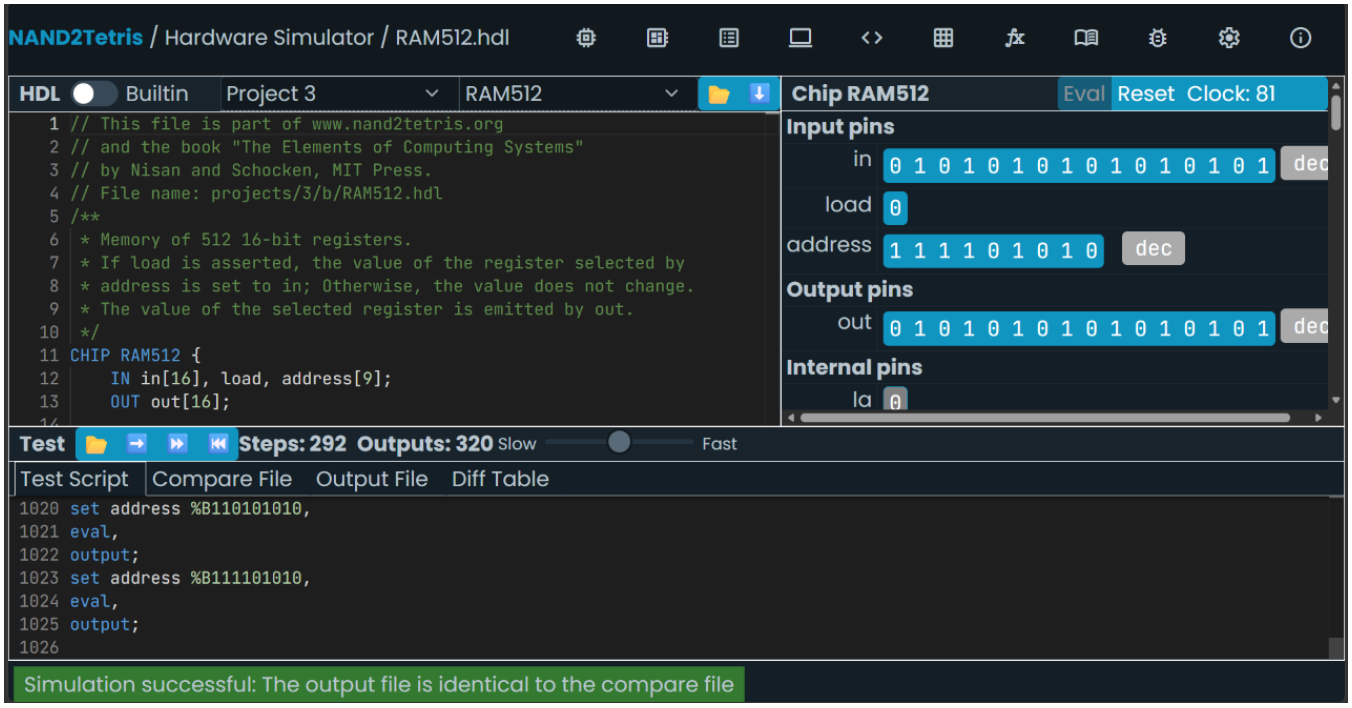
3. RAM8



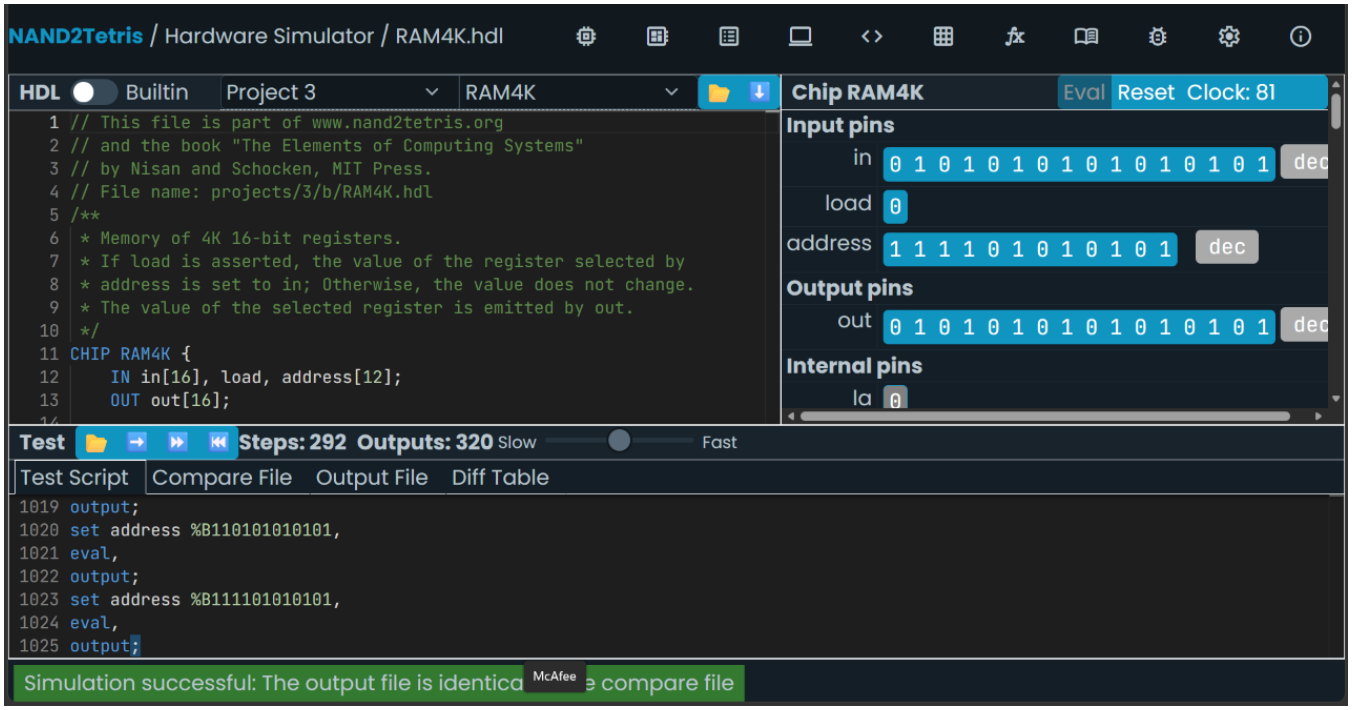
4. RAM64



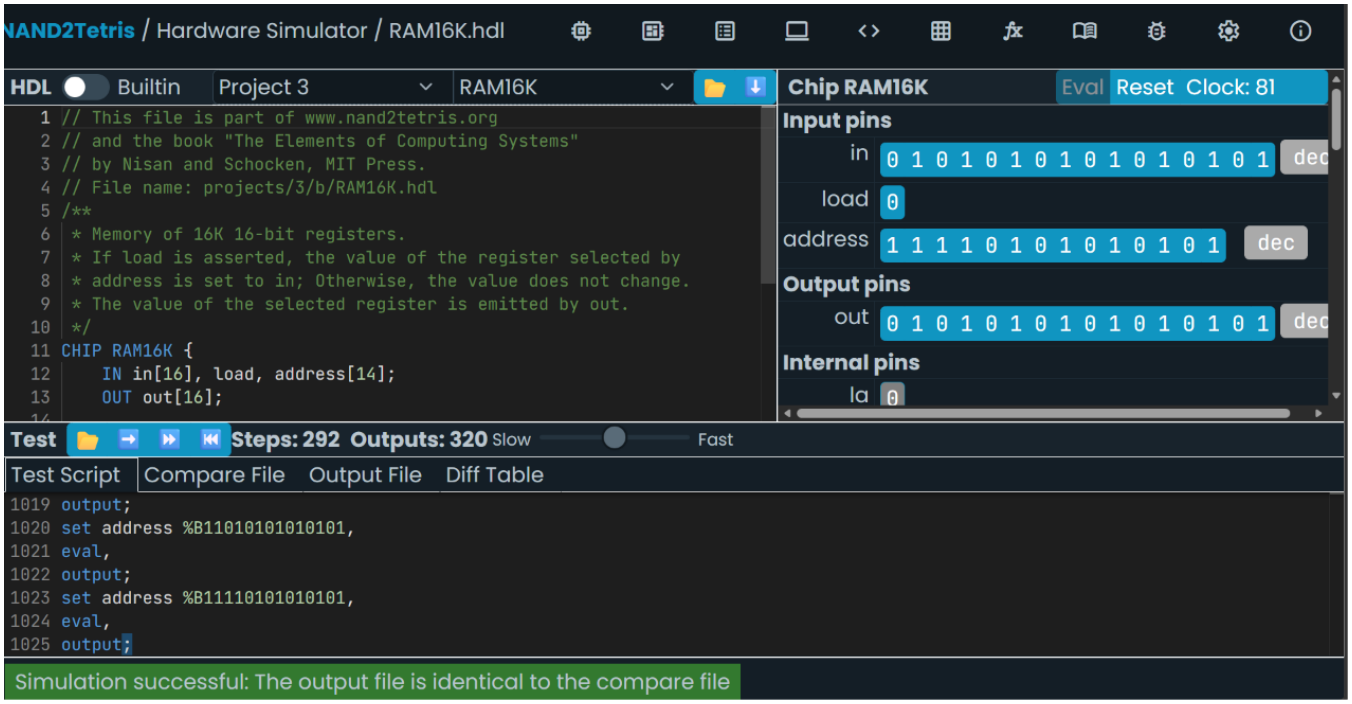
5. RAM512



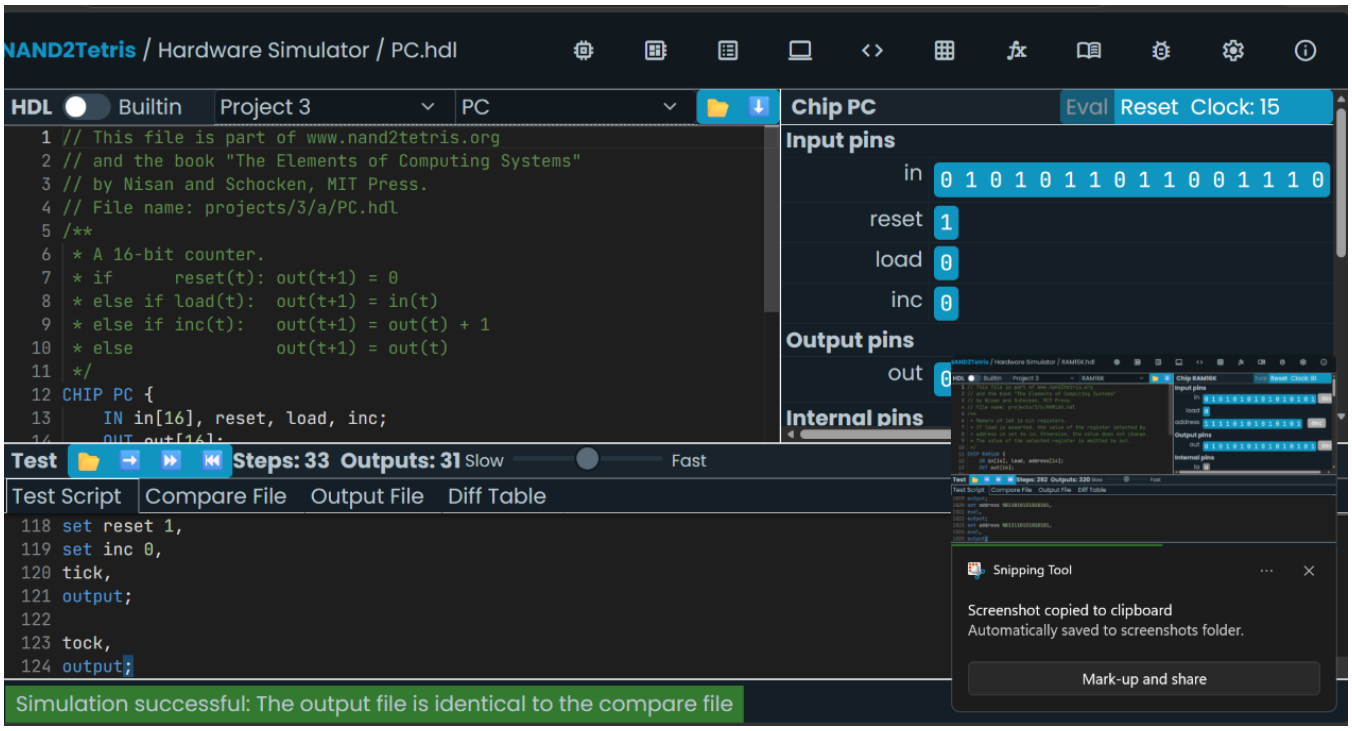
6. RAM4K



7. RAM16K


NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL Builtin Project 3 RAM16K
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.hdl
/**
* Memory of 16K 16-bit registers.
* If load is asserted, the value of the register selected by
* address is set to in; Otherwise, the value does not change.
* The value of the selected register is emitted by out.
*/
CHIP RAM16K {
IN in[16], load, address[14];
OUT out[16];
Chip RAM16K Eval Reset Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 1 0 1 0 1 0 1 0 1 0 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
Test Steps: 292 Outputs: 320 Slow Fast
Test Script Compare File Output File Diff Table
1019 output;
1020 set address %B11010101010101,
1021 eval,
1022 output;
1023 set address %B11110101010101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file

8. PC


NAND2Tetris / Hardware Simulator / PC.hdl
HDL Builtin Project 3 PC
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/PC.hdl
/**
* A 16-bit counter.
* if reset(t): out(t+1) = 0
* else if load(t): out(t+1) = in(t)
* else if inc(t): out(t+1) = out(t) + 1
* else out(t+1) = out(t)
*/
CHIP PC {
IN in[16], reset, load, inc;
Chip PC Eval Reset Clock: 15
Input pins
in 0 1 0 1 0 1 1 0 1 1 0 0 1 1 1 0
reset 1
load 0
inc 0
Output pins
out
Internal pins
Test Steps: 33 Outputs: 31 Slow Fast
Test Script Compare File Output File Diff Table
118 set reset 1,
119 set inc 0,
120 tick,
121 output;
122
123 tock,
124 output;
Simulation successful: The output file is identical to the compare file
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share